6 **(a)** State what is meant by *mass defect*.

.....

.....

..... [1]

(b) **(i)** The mass of the helium−3 nuclide $\left({}^{3}_{2}\mathrm{He}\right)$ is 3.014931 u.

The mass of a proton is 1.00728 u while the mass of a neutron is 1.00867 u.

Show that the binding energy per nucleon for the helium−3 nuclide is 2.6 MeV.

[3]

(ii) The binding energy per nucleon for deuterium $\left({}^{2}_{1}\mathrm{H}\right)$ is 1.11 MeV.

Calculate the energy released in the following reaction:

$${}^{2}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + \gamma$$

energy released = ... MeV [2]

(c) The axes in Fig. 6.1 show the variation with nucleon number *A* with the binding energy per nucleon.

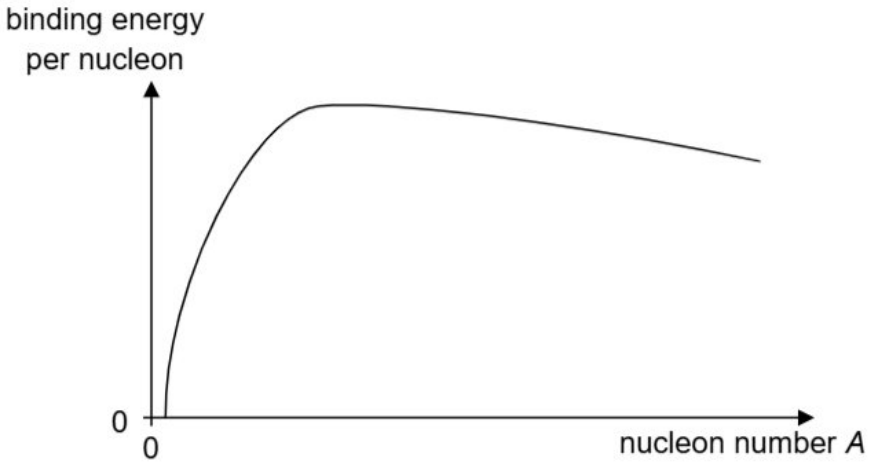


Fig. 6.1

(i) On Fig. 6.1, mark and label two points corresponding to the approximate positions of deuterium $\left({}^{2}_{1}\mathrm{H}\right)$ and helium-3 $\left({}^{3}_{2}\mathrm{He}\right)$.

[1]

(ii) With reference to Fig. 6.1, explain why energy is released in the reaction in **(b)(ii)**.

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.....

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..... [2]